

WM7 details leaked

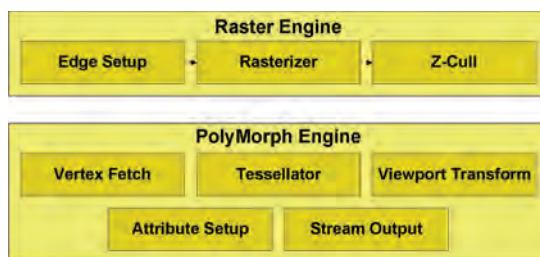
Microsoft is livid at leaked details on the Windows Mobile 7 development front

Facebook at number two

Good old FB is now the second most popular website in the US, behind only Google

Tessellation + Displacement Mapping = visual realism

Both these techniques are not new, and are used extensively in the film industry as animation techniques. When used together, developers have precise control on the level of geometric detail, which is very important. Animation is done on a compact description and this is scaled up to whatever an appropriate quality level. This also brings about savings in memory usage as well as bandwidth consumption. In the example below, courtesy Kenneth Scott, id Software, we see a general outline of the character on the left, note the very limited geometry due to which the object looks rough. Using Tessellation, we see a result free from gradients in the centre, but no more detailed than the object on the left. The object on the right is a result of applying a displacement map to the smoothly tessellated object in the middle and looks much more realistic.



Better AA

On the GF100 enabling 8x MSAA (Multi Sample Anti Aliasing) no longer incurs the performance hit that it was associated with. CSAA (Coverage Sample Anti Aliasing), which is the frugal mans AA technique has also been tweaked with support for up to 32 samples. Transparency Multi-Sample Anti Aliasing incurs less of a performance hit on the GF100 and thanks to more samples, the overall effect is a little close to realistic.

More gamer-goodies

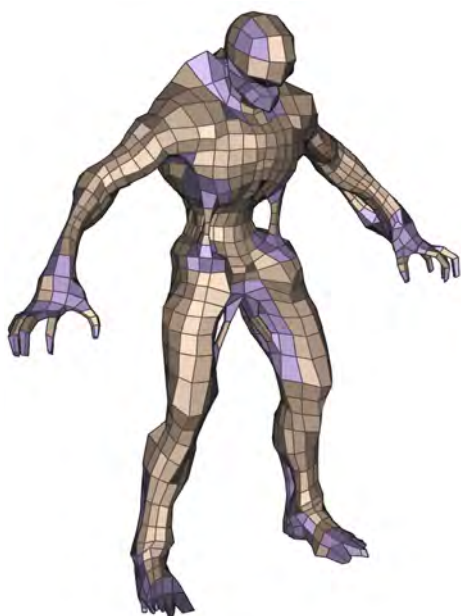
NVIDIA has showcased 3D Vision Surround, which involves their 3D Vision and up to three monitors for an immersive experience. 3D Vision with Batman: Arkham Asylum was

really fun and immersive – the action really seemed to jump out at you. Three monitors with such a 3D effect would be even more amazing. AMD gets kudos for marketing the multiple monitor concept first. Their technology is called Eyefinity, where 2, 3, and up to 6 monitors can be used for a truly larger than life (literally) gaming experience. But NVIDIA has added the 3D angle to this, going a very positive step further.

ECC: For Tesla, with love?

Another much talked about feature is ECC or Error Correction Code. While the RV870 (Radeon HD 5870) can detect errors on the memory bus it cannot correct them. The GF100 sees ECC support on both L1 and L2 cache as well as the register file. Obviously this is a feature included to keep the GF100s chances as a GPGPU alive

ECC is mandatory for simulations and large scale floating point calculations, meaning nobody would touch the GF100 for Tesla-like functions unless ECC was a part of the package. ECC does nothing for games, and one pitfall is while overclocking the memory system could behave weirdly.



Tessellation along with displacement mapping adds additional detail to the mesh